Technical Report: Construction and Canonicalization of the Neomelodic Corpus

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1 Introduction

This report describes the construction pipeline of the *Neomelodic Corpus* and its current state after the June 2026 methodological reorganization. The central point of the update is the separation between a **raw** layer, a **canonical track-analysis** layer, and a **lyrics layer with quality tiers**. This step was necessary because the old setup mixed release duplicates, title variants, and texts with highly heterogeneous quality within the same denominator.

2 Stage 1: Seed and Artist Selection

The identification of the starting sample follows a funnel logic:

- broad collection of neomelodic candidates from discographic, encyclopedic, and sectoral sources;
- cross-source coherence scoring;
- nominal deduplication and final profile validation.

The current analytical artist layer contains **190 core artists**. This layer remains stable and is today the most solid block of the pipeline: the audit did not reveal structural problems comparable to those that emerged for tracks and lyrics.

3 Stage 2: Discography Construction

The full discography is reconstructed by integrating multiple sources, especially Discogs, MusicBrainz, and iTunes, with provenance traces for release and track. The raw reference file remains `canonical_discography.csv`, which contains **45,815** discographic records.

For analysis, however, this raw export cannot be used as the final denominator: many rows represent the same song repeated across albums, compilations, or editorial variants. For this reason, a new canonical title-level dataset was introduced:

- `tracks_final_content.dta`: legacy layer with **36,660** rows and release-level duplication still present;
- `tracks_unique_title_level.dta`: current canonical layer with **32,508** unique artist-title pairs after coherent title normalization.

Title-level deduplication now uses the same normalization employed for lyrics matching. This removes an important ambiguity: the previous version retained many minor orthographic variants of the same track, which then produced multiple merges with a single text.

4 Stage 3: Lyrics Retrieval

Lyrics retrieval still starts from the corpus database, which currently contains **8,815** raw lyric rows associated with tracks. The sources effectively observed in the final layer are:

- `LRCLIB`
- `YouTube_Archive_Strict`
- `YouTube`
- `rescue_surgical`

Retrieval is no longer described as a simple "single gold standard." The audit showed that the available texts have very different quality levels, and treating them as homogeneous obscures both real coverage and analytical reliability.

5 Stage 4: Filtering, Deduplication, and New Canonical Layers

5.1 Filtering Non-Relevant Content

Each retrieved text is subjected to cleaning and relevance filtering:

- removal of biographies, spoken-word material, sketches, and non-song content;
- exclusion of cases linguistically too far from the corpus perimeter;
- text normalization for duplicate comparison.

The process moves from **8,815** raw rows to **7,441** rows that pass the basic relevance filter.

5.2 Lyrics Deduplication

After the basic filter, texts are first reduced to unique candidates by normalized artist-title pair and then subjected to semantic deduplication **only within artist**. This choice replaces the old global cross-artist deduplication, which was too aggressive and methodologically difficult to defend as a default.

The current steps are:

- **4,672** unique title-level candidates by artist;
- **788** cases removed as intra-artist semantic duplicates;
- **3,884** final texts in the new canonical lyrics layer.

5.3 Quality Tiers

The final lyrics layer is no longer treated as a homogeneous block. Texts are now divided into two classes:

- **high_confidence**: **2,601** texts, obtained from more reliable sources or quality signals;
- **rescue**: **1,283** texts, retained for coverage but to be used with greater caution.

Classification uses text provenance and simple structural indicators of content: length, lexical duplication, line repetition, and anomalous token signals.

5.4 Translations

The historical pipeline included an Italian translation column in the old export `lyrics_dataset.csv`. In the new canonical layer, this information is currently maintained only as a legacy inheritance from the old export when available:

- **64.7%** of layered texts have a non-empty inherited Italian translation;
- in **55.9%** of cases the translation field exactly matches the original text, so it should not automatically be interpreted as a true semantic translation.

This point remains open: translation has not yet been re-estimated uniformly on the new canonical layer.

5.5 Traceability and Provenance

Each record in the new canonical layers maintains information useful for provenance:

- normalized artist and title keys;
- text source;
- quality tier;
- exact-text-shared-across-artists flag;
- in the track layer, number of collapsed rows and a summary of source releases.

This structure finally makes explicit the difference between *raw records*, *title-level analytical units*, and *text usable for analysis*.

5.6 Consolidated Statistics

The canonical datasets recommended after the audit are:

- `artists_individual.dta`: **190** artists;
- `tracks_unique_title_level.dta`: **32,508** canonical title-level tracks;
- `lyrics_dataset_layered.dta`: **3,884** canonical texts;
- `lyrics_analysis_layered.dta`: **3,884** coherent one-to-one matches between canonical tracks and lyrics.

Lyrics coverage over the canonical track denominator is therefore **11.95%**. Restricting to *high_confidence* texts only, coverage drops to **8.00%**. This is a methodologically more honest result than the old count based on a denominator that still contained duplicates.

Artist	**Canonical Tracks**	**Total Lyrics**	**High Confidence**
Enzo di Domenico	1427	209	19
Gigi D'Alessio	265	186	168
Gianni Vezzosi	449	158	124
Gigi Finizio	243	144	118
Gianni Fiorellino	535	142	108
Mario Merola	1484	123	40
Gianni Celeste	273	115	103
Tony Colombo	158	107	97
Antonio Buonomo	550	97	7
Sal Da Vinci	175	96	87
Mauro Nardi	1057	93	33
Gianluca	138	79	79
Canonical corpus total	**32,508**	**3,884**	**2,601**

Methodological note: The table uses the new denominator `tracks_unique_title_level.dta` and the new layer `lyrics_dataset_layered.dta`. Lyrics counts are therefore title-level rather than release-level.

6 Conclusions

The corpus should no longer be described as a single final dataset that is perfectly homogeneous. The correct description today is this:

- there is a very rich raw layer, but it is not directly analytical;
- there is a credible canonical track layer at the title-artist level;
- there is a canonical lyrics layer that separates coverage from reliability;
- the main residual weakness is no longer deduplication, but still-limited lyrics coverage and the non-uniform quality of the *rescue* block.

In this form, the *Neomelodic Corpus* is more scientifically defensible: less ambiguous about the denominator, more transparent about provenance, and more readable regarding the real conditions under which lyrics are used in downstream reports.